MasterCinator

Technical Design & Architectural Specification

Project: MasterCinator Video Captioning Agent

Target Model: MiniMax M3 (Fireworks AI)

Target Platform: linux/amd64

Architecture: Two-Stage VLM Critique Pipeline

1. Executive Summary

MasterCinator is a high-performance, single-file deployment agent engineered for advanced multimodal video understanding. Designed specifically to ingest video tasks and output highly accurate captions across four creative personas, it prioritizes a minimal footprint, security isolation, and architectural robustness.

Unlike standard visual captioning systems that rely on naive single-shot models, MasterCinator implements a novel two-stage visual critique pipeline. By partitioning the video timeline into alternate sets of chronological keyframes, the system generates styled drafts and subsequently cross-references them against independent verification frames to filter out hallucinations.

2. Pipeline Architecture

The captioning execution pipeline runs as a self-contained flow inside a Docker container. The flow of a single video task is structured as follows:

2.1 Temporal Sampling and Pre-processing

The agent retrieves the target video URL and downloads the raw clip. OpenCV is used to temporally sample exactly 8 keyframes spanning the entire video duration. Each keyframe is resized such that the longest side is 1024px. This preserves visual details like text overlays and tiny subjects while keeping the API upload payload lightweight.

2.2 Stage 1: Multimodal Persona Generation (VLM Call 1)

The extracted frames are divided into alternate sets. The first set contains frames at indices **[0, 2, 4, 6]**. These are base64-encoded and sent to MiniMax M3. In a single multimodal query, the model is instructed to write four captions (Formal, Sarcastic, Tech Humorous, Non-Tech Humorous) using our detailed few-shot styling guidelines, returning a single JSON object.

2.3 Stage 2: Self-Critique Verification (VLM Call 2)

To guarantee factual grounding, the remaining keyframes at indices **[1, 3, 5, 7]** (representing intermediate segments of the video timeline) are packaged. The agent sends these verification frames along with the Stage 1 draft JSON to MiniMax M3. The model reviews the draft captions against the new frames to identify hallucinations, brand names, or visual overlays, correcting the text as necessary. The final verified JSON block is written to the output file.

3. Technical Specifications & Stack

The project is packaged to be fully self-contained, using standard system configurations and lightweight python dependencies:

Specification	Configuration Details
Base Image	python:3.12-slim
System Dependencies	ffmpeg, curl
Python Libraries	openai, opencv-python-headless, requests, python-dotenv, streamlit
VLM Model	accounts/fireworks/models/minimax-m3
Build Target	linux/amd64 (cross-compiled using Docker buildx)
Runtime Interface	agent.py (File-to-file processing via /input and /output directories)
Interactive Interface	demo.py (Streamlit UI for direct video upload analysis)

4. Prompt Persona Specifications

MasterCinator enforces strict stylistic personas that are injected into Call 1 and checked in Call 2:

- **Formal:** Objective, factual, and neutral, in the register of a documentary narrator. No humor, opinions, or exclamations.
- **Sarcastic:** Dry, ironic, deadpan, and lightly mocking, as if gently unimpressed. Keep the humor grounded in the actual scene.
- **Humorous Tech:** A funny caption for a developer audience using ONE tech metaphor. Build the whole joke around it.
- **Humorous Non-Tech:** Warm, relatable, everyday observational humor. Do NOT use any programming or technical jargon.